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GitHub Link	https://github.com/YOGAMADHA/MLA0405-DEEP-LEARNING

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL 1
Case Study

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0405**
SLOT : D

COURSE OUTCOME COVERED

CO3: Analyze and apply convolutional neural network architectures, including convolutional layers, filters, parameter sharing, regularization, AlexNet and ResNet, to develop solutions for image classification and real-world computer vision applications. (L4)

CO Weightage: 25%
Assessment Tool Weightage: 25%

Duration: 90 minutes
Marks: 25

Code of Conduct:

I **<u>Yoga Madha A-192425058</u>** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Yogamadha

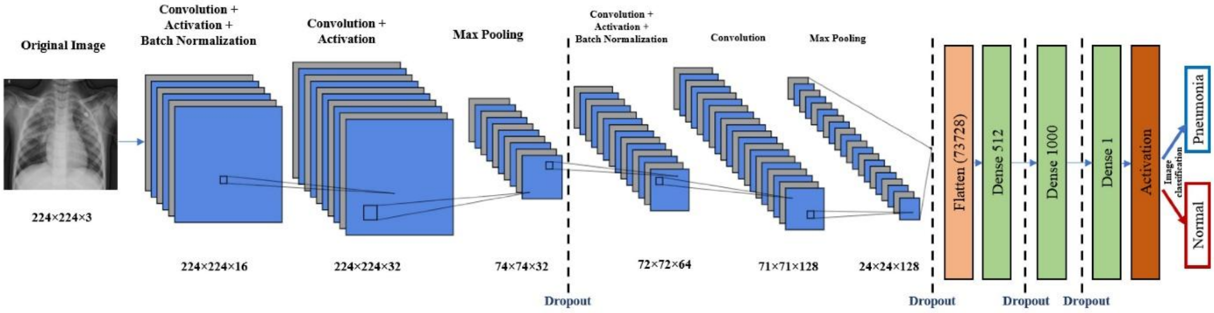
Signature of the student **Name of the student: Yoga Madha**

Criteria	Conceptual Understanding (2.5)	Linkages (2.5)	Organization (2)	Integration of Literature (2)	Clarity & Presentation *(1)*	Total (10)
Weightage	*25 %*	*25%*	*20%*	*20 %*	*10 %*	100%
Q1						
Q2						
Total						

Signature of the Faculty: ***Total Marks :***

QUESTIONS:02 **Total Marks: 20**

Case Study Question 1: CNN for Automated Medical Image Classification
A healthcare organization wants to develop a deep learning system to automatically classify chest X-ray images into **normal** and **disease-affected** categories. The dataset contains thousands of X-ray images with variations in size, brightness and image quality. A CNN model is proposed because it can automatically learn important visual features such as edges, textures, and abnormal regions without manually designing features.
Question:
1. Design and explain a suitable CNN architecture for this medical image classification system.
Answer:
A suitable CNN architecture for chest X-ray classification can be designed as:
Input Image → Convolution → ReLU → Max Pooling → Convolution → ReLU → Max Pooling → Convolution → ReLU → Max Pooling → Flatten → Fully Connected → Dropout → Output



The chest X-ray images are first **resized to 224 × 224 pixels** and normalized to handle differences in image size and brightness. The first convolutional layer can use **32 filters of size 3×3** to learn basic features such as edges and intensity changes. A **2×2 max-pooling layer** reduces the spatial dimensions.
The second convolutional layer can use **64 filters** to learn textures and shapes. Another pooling layer reduces the feature-map size. A third convolutional layer with **128 filters** can learn more complex anatomical patterns and possible disease-related abnormalities.
The extracted feature maps are then **flattened** and passed to a fully connected layer with around **128 neurons**. Dropout can be applied to reduce overfitting. Finally, a **sigmoid output neuron** classifies the image as either **Normal** or **Disease-Affected**.

2. Describe the motivation for using CNNs and explain the role of convolutional layers, filters, pooling layers, and fully connected layers.
Answer:
CNNs are suitable because they can **automatically learn important visual features directly from X-ray images** without manually designing features. They preserve spatial information and are effective for detecting patterns such as edges, textures, shapes, and abnormal regions.

- **Convolutional layers:** Extract features from the X-ray. Early layers detect simple features such as edges and lines, while deeper layers learn complex anatomical and disease-related patterns.
- **Filters:** Small learnable matrices, such as 3×3 filters, that move across the image and detect specific patterns. Different filters learn different features.
- **Pooling layers:** Reduce the spatial dimensions of feature maps while retaining important information. Max pooling keeps the strongest activation and reduces computational cost.

- **Fully connected layers:** Combine the extracted high-level features and use them to make the final classification between normal and disease-affected images.

Thus, CNN feature learning progresses from:
Edges → Textures → Shapes → Anatomical structures → Abnormal patterns → Classification

3. Explain how parameter sharing reduces the number of trainable parameters compared with a fully connected network.
Answer:
Parameter sharing means that the **same convolutional filter is reused at different locations of the image**.
For example, a 3×3 filter contains only:

$$3\times3=9$$

weights, plus one bias. The same 9 weights are applied across the entire X-ray image.
In contrast, a fully connected network requires separate weights between every input pixel and every neuron. For a **224 × 224** grayscale image:

$$224\times224=50{,}176$$

input pixels are present. Therefore, one fully connected neuron would require approximately **50,176 weights**, excluding bias.
Hence, CNNs use far fewer trainable parameters because the same filter is shared across the image.
This:

- Reduces memory usage
- Reduces computational complexity
- Makes training faster
- Reduces overfitting
- Allows the same feature to be detected at different locations

4. Discuss suitable regularization techniques such as dropout, data augmentation and batch normalization to reduce overfitting.
Answer:
Dropout: Randomly deactivates a proportion of neurons during training. For example, a dropout rate of 0.5 temporarily removes approximately half of the selected neurons during each training step. This prevents the model from depending too much on particular neurons and improves generalization.
Data Augmentation: Creates additional variations of training X-rays using suitable transformations such as small rotations, scaling, cropping, brightness and contrast changes. This helps the CNN handle variations in real X-ray images and reduces overfitting. Medical transformations should remain realistic so that anatomical information is not distorted.
Batch Normalization: Normalizes intermediate activations during training. It helps stabilize and speed up training and can also provide a regularization effect.
Therefore, using **data augmentation + dropout + batch normalization** can help the CNN learn general disease-related features instead of memorizing the training images.

5. Explain whether AlexNet or ResNet would be more appropriate and justify your choice. ResNet would be more appropriate for this application.
Answer:

Factor	AlexNet	ResNet
Network depth	Relatively shallow	Much deeper
Feature learning	Good	Strong

Factor	AlexNet	ResNet
Skip connections	No	Yes
Complex medical features	Moderate	Better
Training deep networks	More difficult	Easier
Computational cost	Lower	Higher depending on version
Classification capability	Good	Generally better

AlexNet is an earlier CNN architecture with relatively fewer layers. It can learn useful image features and requires comparatively less computation, but it has limited depth for learning highly complex patterns.

ResNet uses residual or skip connections:

$$Output = F(x) + x$$

These connections help information and gradients flow through deep networks, reducing problems associated with training very deep CNNs. This allows ResNet to learn more complex hierarchical features from chest X-rays.

For this application, subtle abnormalities may require deeper feature representations. Therefore, **ResNet is preferred for better feature learning and classification performance**. A smaller model such as **ResNet-18 or ResNet-34** can be selected when computational resources are limited, while deeper variants can be used when sufficient computing resources are available.

Case Study Question 2: CNN-Based Autonomous Vehicle Object Recognition (10 Marks)

An autonomous vehicle company is developing a vision system that must identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from images captured by cameras mounted on the vehicle. The system must recognize objects under different lighting conditions, viewing angles, distances and road environments. The development team is considering **AlexNet and ResNet** architectures for building the CNN-based recognition system.

Question:

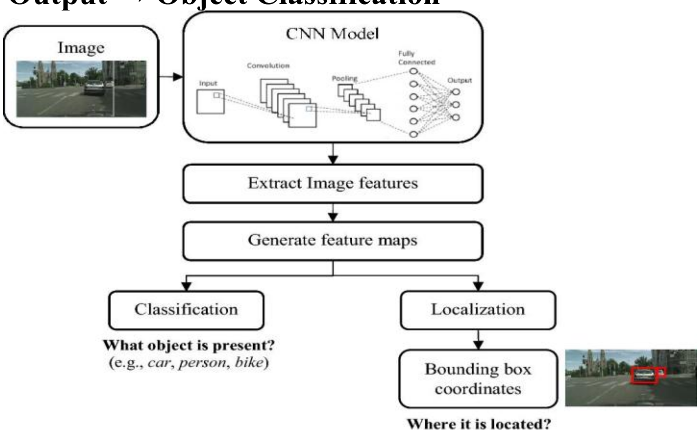
1. Analyze how a CNN architecture can be designed for this autonomous vehicle application.

Answer:

A CNN-based vision system can be designed to identify **cars, pedestrians, traffic signs, bicycles, and road obstacles** from camera images. Since the vehicle operates under different lighting, viewing angles, distances, and road conditions, the CNN should learn both simple and complex visual features.

A suitable architecture is:

Camera Image → Preprocessing → Convolution + ReLU → Pooling → Convolution + ReLU → Pooling → Deeper Convolution Layers → Feature Extraction → Fully Connected/Detection Output → Object Classification



The input images are resized and normalized before being given to the CNN. Early convolutional layers learn basic features such as edges, corners, and textures. Deeper layers combine these features to recognize object shapes and complete objects.
For an actual autonomous vehicle, an **object-detection CNN** is preferable to simple image classification because the system needs to identify **what the object is and where it is located**. The output can therefore contain the object class, bounding box, and confidence score.
For example:
Car → 96% confidence → Bounding box
Pedestrian → 91% confidence → Bounding box
The architecture should also be optimized for fast inference because the vehicle needs to make decisions in real time.

2. Explain the purpose of the input, convolution, activation, pooling, and fully connected/output layers and describe how convolutional filters progressively learn low-level and high-level features.
Answer:
Input Layer
The input layer receives images captured by the vehicle's cameras. The images are resized to a standard resolution and normalized.
The system must handle:

- Day and night images
- Bright and dark conditions
- Different viewing angles
- Near and distant objects
- Different road environments

Convolutional Layers
Convolutional layers extract useful visual features using learnable filters.
The features are learned progressively:
Low-level features → Mid-level features → High-level features
Early layers learn:

- Edges
- Lines
- Corners
- Simple textures

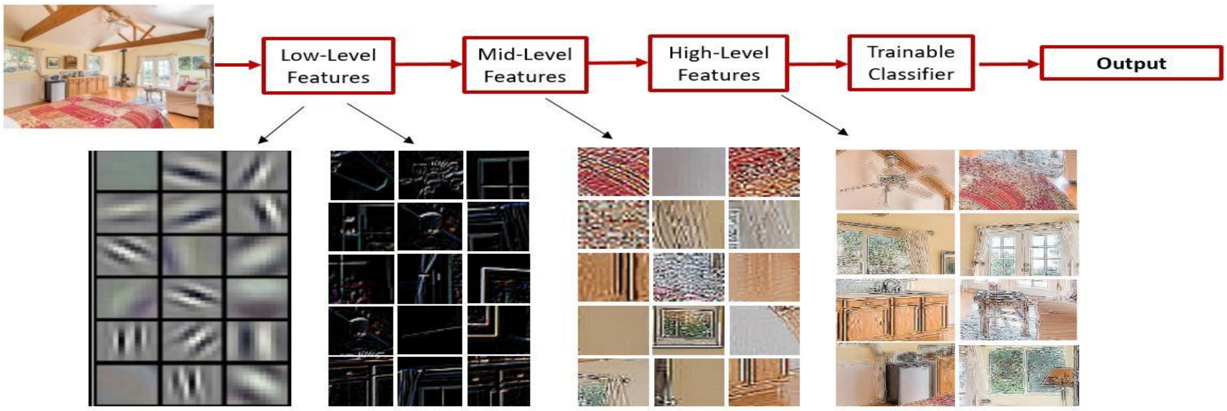
Middle layers learn:

- Shapes
- Object parts
- Wheels
- Human body patterns
- Traffic-sign patterns

Deep layers learn:

- Cars
- Pedestrians
- Bicycles
- Traffic signs
- Road obstacles

- Input image pixels → Edges → Textures → Parts → Objects



Activation Layer

The ReLU activation function is commonly used:

$$\text{ReLU}(x)=\max(0,x)$$

It introduces non-linearity and allows the network to learn complex patterns instead of only simple linear relationships.

Pooling Layer

Pooling reduces the spatial dimensions of feature maps.

For example, **2×2 max pooling** selects the largest value from each region. This reduces computation and retains strong features.

Pooling also provides some robustness when an object moves slightly within the image.

Fully Connected/Output Layer

After feature extraction, the learned features are combined to determine the final object class.

For an object-recognition system, the output can provide:

- **Object class**
- **Location/bounding box**
- **Confidence score**

Thus, the overall learning process is:

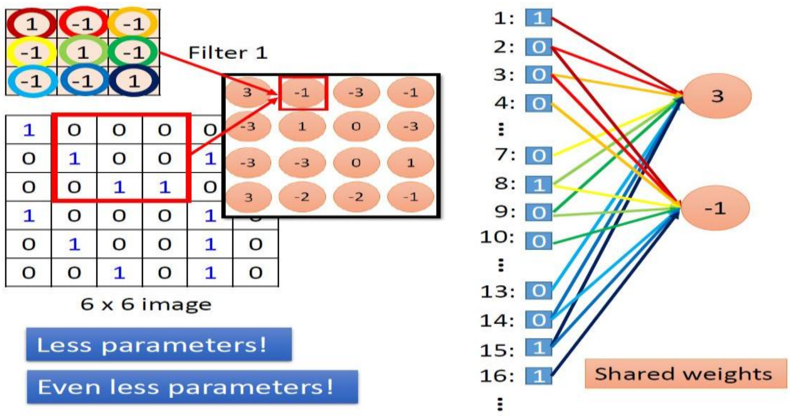
Pixels → Edges → Textures → Shapes → Object parts → Complete objects

3. Explain the importance of parameter sharing in reducing computational complexity and improving translation-related feature detection.

Answer:

Parameter sharing means that the **same convolutional filter is used at different locations of the image**.

For example, a filter that learns to detect a vertical edge can be reused throughout the camera image.



Without parameter sharing, a separate set of weights would be required for different image locations. This would produce a very large number of parameters.
With parameter sharing:
One filter → reused across the entire image
Therefore, parameter sharing:

- Reduces the number of trainable parameters
- Reduces memory requirements
- Reduces computational complexity
- Makes CNN training more efficient
- Allows the same feature to be detected at different positions

This is particularly important for autonomous vehicles because an object may appear anywhere in the camera's field of view.
For example, a **traffic sign** may appear on the left side of one image and the right side of another. The same learned filters can detect its visual characteristics in both locations.

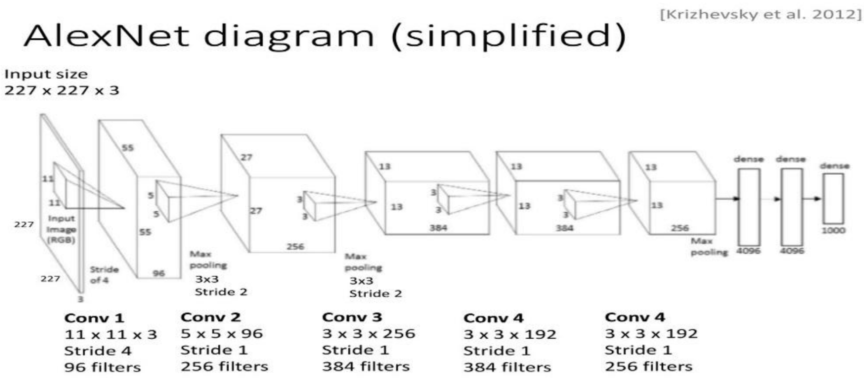
4. Discuss the possible problem of overfitting and recommend suitable regularization methods.
Answer:
Overfitting occurs when the CNN performs very well on training images but performs poorly on new road scenes.
For example, the model might memorize specific roads, vehicles, or backgrounds instead of learning general object characteristics.
Data Augmentation
Training images can be modified using:

- Small rotations
- Scaling
- Cropping
- Brightness changes
- Contrast changes
- Blur
- Noise
- Different weather simulations

This helps the model recognize objects under different real-world conditions.
Dropout
Dropout randomly disables some neurons during training. This prevents the network from depending too strongly on particular neurons and improves generalization.
Batch Normalization
Batch normalization normalizes intermediate activations during training. It helps stabilize training and can provide a regularization effect.
L2 Regularization
L2 regularization penalizes excessively large weights and encourages the model to learn a more general representation.
Early Stopping
Training can be stopped when validation performance stops improving. This prevents the model from continuing to memorize the training dataset.
Therefore, a combination of **data augmentation, dropout, batch normalization, L2 regularization, and early stopping** can reduce overfitting.

5. Compare AlexNet and ResNet in terms of architecture, depth, parameter efficiency, training difficulty, vanishing-gradient problems and feature-learning capability.

Answer:



Feature	AlexNet	ResNet
Architecture	Conventional CNN	Deep CNN with residual/skip connections
Depth	Relatively shallow	Available in 18, 34, 50, 101, 152+ layers
Parameter efficiency	Less efficient for very deep feature extraction	Residual design allows deeper networks without simply relying on sequential layers
Training difficulty	Easier because it is shallower	Deep networks are easier to optimize because of skip connections
Vanishing-gradient problem	More difficult to handle as depth increases	Skip connections improve gradient flow
Feature learning	Good for basic image features	Strong for complex hierarchical features
Computational cost	Relatively lower	Higher depending on ResNet version
Complex road scenes	Moderate capability	Better capability
Suitability	Basic recognition/baseline	Better for complex autonomous-vision tasks

AlexNet

AlexNet is an earlier CNN architecture with a comparatively shallow structure. It can successfully learn edges, textures, and shapes, but its feature-learning capability is more limited compared with modern deep architectures.

ResNet

ResNet introduces **residual connections** that allow information to bypass some convolutional layers.

The basic relationship is:

$$y=F(x)+x$$

The shortcut connection helps gradients flow through the network and makes deep feature learning easier.

ResNet can therefore learn more complex representations needed for recognizing objects under different scales, angles, and environments.

6. Recommend the most suitable architecture and justify your decision based on accuracy, computational cost and real-time requirements.

Answer:

ResNet is the more suitable architecture for the autonomous vehicle system.
The main reason is that autonomous vehicles must recognize objects under highly variable conditions. ResNet can learn deeper and more complex features than AlexNet.

Accuracy

ResNet's deeper architecture can provide stronger feature representations for:

- Small pedestrians
- Distant vehicles
- Traffic signs
- Bicycles
- Complex road obstacles

Therefore, ResNet is generally preferable when recognition accuracy is important.

Computational Cost

A deeper ResNet requires more computational resources. However, the solution is not necessarily to use the deepest possible ResNet.

A suitable model can be selected according to available hardware:

Limited resources → ResNet-18 / ResNet-34

More powerful GPU/accelerator → ResNet-50 or deeper variant

Real-Time Requirements

Autonomous vehicles require fast inference. A highly accurate model that is too slow may not be practical.

Therefore, the model should provide a balance between:

Accuracy + Inference Speed + Memory Usage + Computational Cost

A relatively lightweight ResNet can provide a good balance between strong feature learning and real-time processing.

Final Recommendation

ResNet is recommended over AlexNet because it provides stronger feature learning, supports deeper networks, handles gradient-flow problems better through residual connections, and is more suitable for complex road-scene recognition. For real-time autonomous driving, a smaller ResNet such as **ResNet-18 or ResNet-34** can be selected when computational resources and response time are important.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand